

# Linear Regulator - Wide Input Voltage Range, Ultra-Low Iq, High PSRR 10 mA

## NCP785A

The NCP785A is a high-performance linear regulator, offering a very wide operating input voltage range of up to 450 V DC, with an output current of up to 10 mA.

Ideal for high input voltage applications such as industrial and home metering, home appliances. The NCP785A family offers  $\pm 5\%$  initial accuracy, extremely high-power supply rejection ratio and ultra-low quiescent current. The NCP785A is optimized for high-voltage line and load transients, making this part ideal for harsh environment applications.

The NCP785A is offered in fixed output voltage options 3.3 V, 5.0 V, 12 V and 15 V.

SOT-89 package offers good thermal performance and help to minimize the solution size.

### Features

- Wide Input Voltage Range:  
DC: Up to 450 V  
AC: 85 V to 260 V (half-wave rectifier and 2.2  $\mu$ F capacitor)
- 10 mA Guaranteed Output Current
- Ultra Low Quiescent Current: Typ. 10  $\mu$ A ( $V_{OUT} \leq 5$  V)
- $\pm 5\%$  Accuracy Over Full Load, Line and Temperature Variations
- Ultra-high PSRR: 70 dB at 60 Hz, 90 dB at 100 kHz
- Stable with Ceramic Output Capacitor 22  $\mu$ F MLCC
- Thermal Shutdown and Current Limit Protection
- Available in Thermally Enhanced SOT89-3 Package
- This is a Pb-Free Device

### Typical Applications

- Industrial Applications, Home Appliances
- Home Metering / Network Application
- Off-line Power Supplies

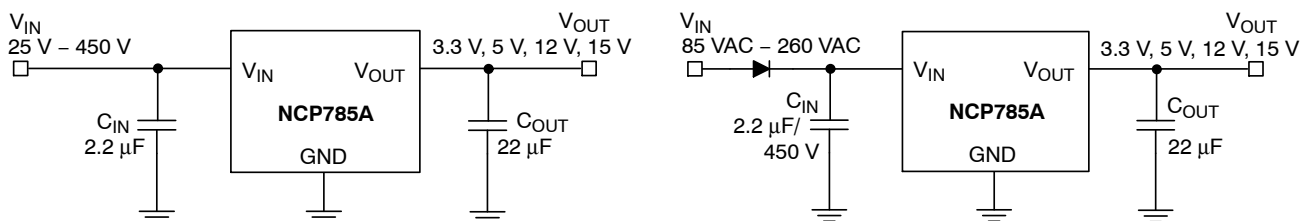
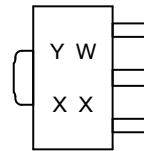


Figure 1. Typical Applications



SOT-89  
CASE 528AG

### MARKING DIAGRAM



Y = Year  
W = Work Week  
XX = Specific Device Code

### ORDERING INFORMATION

See detailed ordering and shipping information in the package dimensions section on page 9 of this data sheet.

## NCP785A

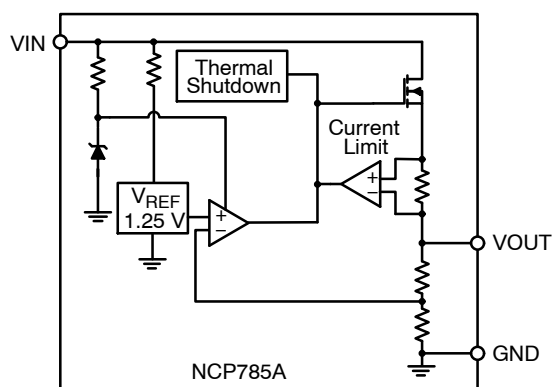


Figure 2. Simplified Internal Block Diagram

Table 1. PIN FUNCTION DESCRIPTION

| Pin No.<br>(SOT-89) | Pin Name | Description                                                                     |
|---------------------|----------|---------------------------------------------------------------------------------|
| 1                   | VIN      | Supply Voltage Input. Connect 2.2 $\mu$ F capacitor from VIN to GND.            |
| 2, Tab              | GND      | Ground connection.                                                              |
| 3                   | VOUT     | Regulator Output. Connect 22 $\mu$ F or larger MLCC capacitor from VOUT to GND. |

Table 2. ABSOLUTE MAXIMUM RATINGS

| Rating                                                                  | Symbol       | Value       | Unit         |
|-------------------------------------------------------------------------|--------------|-------------|--------------|
| Input Voltage (Note 1)                                                  | $V_{IN}$     | -0.3 to 700 | V            |
| Output Voltage                                                          | $V_{OUT}$    | -0.3 to 18  | V            |
| Maximum Junction Temperature                                            | $T_{J(MAX)}$ | 150         | $^{\circ}$ C |
| Storage Temperature                                                     | $T_{STG}$    | -55 to 150  | $^{\circ}$ C |
| ESD Capability, Human Body Model (All pins except HV pin no.1) (Note 2) | $ESD_{HBM}$  | 2000        | V            |
| ESD Capability, Machine Model (Note 2)                                  | $ESD_{MM}$   | 200         | V            |

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. Peak 650 V max 1 ms non repeated for 1 s
2. This device series incorporates ESD protection and is tested by the following methods:  
 ESD Human Body Model tested per AEC-Q100-002 (EIA/JESD22-A114)  
 ESD Machine Model tested per AEC-Q100-003 (EIA/JESD22-A115)  
 Latch-up Current Maximum Rating tested per JEDEC standard: JESD78.

Table 3. THERMAL CHARACTERISTICS

| Rating                                                                 | Symbol          | Value | Unit           |
|------------------------------------------------------------------------|-----------------|-------|----------------|
| Thermal Characteristics, SOT-89<br>Thermal Resistance, Junction-to-Air | $R_{\theta JA}$ | 79    | $^{\circ}$ C/W |

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**Table 4. ELECTRICAL CHARACTERISTICS,  $V_{OUT} = 3.3\text{ V}$**  ( $-40^{\circ}\text{C} \leq T_J \leq 85^{\circ}\text{C}$ ;  $V_{IN} = 340\text{ V}$ ;  $I_{OUT} = 100\text{ }\mu\text{A}$ ,  $C_{IN} = 2.2\text{ }\mu\text{F}$ ,  $C_{OUT} = 22\text{ }\mu\text{F}$ , unless otherwise noted. Typical values are at  $T_J = +25^{\circ}\text{C}$ .) (Note 3)

| Parameter                             | Test Conditions                                                                                                                           | Symbol              | Min    | Typ | Max    | Unit                |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|-----|--------|---------------------|
| Operating Input Voltage DC            |                                                                                                                                           | $V_{IN}$            | 25     |     | 450    | V                   |
| Output Voltage Accuracy               | $T_J = 25^{\circ}\text{C}$ , $I_{OUT} = 100\text{ }\mu\text{A}$ , $25\text{ V} \leq V_{IN} \leq 450\text{ V}$                             | $V_{OUT}$           | 3.1515 | 3.3 | 3.4485 | V                   |
|                                       | $-40^{\circ}\text{C} \leq T_J \leq 85^{\circ}\text{C}$ , $I_{OUT} = 100\text{ }\mu\text{A}$ , $25\text{ V} \leq V_{IN} \leq 450\text{ V}$ | $V_{OUT}$           | 3.135  | 3.3 | 3.465  | V                   |
| Line Regulation                       | $25\text{ V} \leq V_{IN} \leq 450\text{ V}$ , $I_{OUT} = 100\text{ }\mu\text{A}$                                                          | $\text{Reg}_{LINE}$ | -0.5   | 0.2 | +0.5   | %                   |
| Load Regulation                       | $100\text{ }\mu\text{A} \leq I_{OUT} \leq 10\text{ mA}$ , $V_{IN} = 35\text{ V}$                                                          | $\text{Reg}_{LOAD}$ | -1.0   | 0.6 | +1.0   | %                   |
| Maximum Output Current (Note 4)       | $35\text{ V} \leq V_{IN} \leq 450\text{ V}$                                                                                               | $I_{OUT}$           | 10.5   |     |        | mA                  |
| Quiescent Current                     | $I_{OUT} = 0$ , $25\text{ V} \leq V_{IN} \leq 450\text{ V}$                                                                               | $I_Q$               |        | 7.5 | 14     | $\mu\text{A}$       |
| Ground Current (Note 4)               | $25\text{ V} \leq V_{IN} \leq 450\text{ V}$<br>$0 < I_{OUT} \leq 10\text{ mA}$                                                            | $I_{GND}$           |        |     | 15     | $\mu\text{A}$       |
| Power Supply Rejection Ratio          | $V_{IN} = 340\text{ V}_{DC} + 1\text{ V}_{pp}$ modulation, $I_{OUT} = 100\text{ }\mu\text{A}$                                             | $f = 1\text{ kHz}$  | PSRR   |     | 70     | dB                  |
| Noise                                 | $f = 100\text{ Hz to } 100\text{ kHz}$<br>$V_{IN} = 340\text{ V}_{DC}$ , $I_{OUT} = 100\text{ }\mu\text{A}$                               | $V_{NOISE}$         |        | 240 |        | $\mu\text{V}_{rms}$ |
| Thermal Shutdown Temperature (Note 5) | Temperature increasing from $T_J = +25^{\circ}\text{C}$                                                                                   | $T_{SD}$            |        | 145 |        | $^{\circ}\text{C}$  |
| Thermal Shutdown Hysteresis (Note 5)  | Temperature falling from $T_{SD}$                                                                                                         | $T_{SDH}$           | -      | 10  | -      | $^{\circ}\text{C}$  |

**Table 5. ELECTRICAL CHARACTERISTICS,  $V_{OUT} = 5.0\text{ V}$**  ( $-40^{\circ}\text{C} \leq T_J \leq 85^{\circ}\text{C}$ ;  $V_{IN} = 340\text{ V}$ ;  $I_{OUT} = 100\text{ }\mu\text{A}$ ,  $C_{IN} = 2.2\text{ }\mu\text{F}$ ,  $C_{OUT} = 22\text{ }\mu\text{F}$ , unless otherwise noted. Typical values are at  $T_J = +25^{\circ}\text{C}$ .) (Note 3)

| Parameter                             | Test Conditions                                                                                                                           | Symbol              | Min   | Typ  | Max   | Unit                |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------|------|-------|---------------------|
| Operating Input Voltage DC            |                                                                                                                                           | $V_{IN}$            | 50    |      | 450   | V                   |
| Output Voltage Accuracy               | $T_J = 25^{\circ}\text{C}$ , $I_{OUT} = 100\text{ }\mu\text{A}$ , $50\text{ V} \leq V_{IN} \leq 450\text{ V}$                             | $V_{OUT}$           | 4.775 | 5.0  | 5.225 | V                   |
|                                       | $-40^{\circ}\text{C} \leq T_J \leq 85^{\circ}\text{C}$ , $I_{OUT} = 100\text{ }\mu\text{A}$ , $50\text{ V} \leq V_{IN} \leq 450\text{ V}$ | $V_{OUT}$           | 4.75  | 5.0  | 5.25  | V                   |
| Line Regulation                       | $50\text{ V} \leq V_{IN} \leq 450\text{ V}$ , $I_{OUT} = 100\text{ }\mu\text{A}$                                                          | $\text{Reg}_{LINE}$ | -0.5  | 0.2  | +0.5  | %                   |
| Load Regulation                       | $100\text{ }\mu\text{A} \leq I_{OUT} \leq 10\text{ mA}$ , $V_{IN} = 55\text{ V}$                                                          | $\text{Reg}_{LOAD}$ | -1.0  | 0.62 | +1.0  | %                   |
| Maximum Output Current (Note 4)       | $55\text{ V} \leq V_{IN} \leq 450\text{ V}$                                                                                               | $I_{OUT}$           | 10.5  |      |       | mA                  |
| Quiescent Current                     | $I_{OUT} = 0$ , $50\text{ V} \leq V_{IN} \leq 450\text{ V}$                                                                               | $I_Q$               |       | 16   | 21    | $\mu\text{A}$       |
| Ground Current (Note 4)               | $50\text{ V} \leq V_{IN} \leq 450\text{ V}$<br>$0 < I_{OUT} \leq 10\text{ mA}$                                                            | $I_{GND}$           |       |      | 23    | $\mu\text{A}$       |
| Power Supply Rejection Ratio          | $V_{IN} = 340\text{ V}_{DC} + 1\text{ V}_{pp}$ modulation, $I_{OUT} = 100\text{ }\mu\text{A}$                                             | $f = 1\text{ kHz}$  | PSRR  |      | 70    | dB                  |
| Noise                                 | $f = 100\text{ Hz to } 100\text{ kHz}$<br>$V_{IN} = 340\text{ V}_{DC}$ , $I_{OUT} = 100\text{ }\mu\text{A}$                               | $V_{NOISE}$         |       | 300  |       | $\mu\text{V}_{rms}$ |
| Thermal Shutdown Temperature (Note 5) | Temperature increasing from $T_J = +25^{\circ}\text{C}$                                                                                   | $T_{SD}$            |       | 145  |       | $^{\circ}\text{C}$  |
| Thermal Shutdown Hysteresis (Note 5)  | Temperature falling from $T_{SD}$                                                                                                         | $T_{SDH}$           | -     | 10   | -     | $^{\circ}\text{C}$  |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

- Performance guaranteed over the indicated operating temperature range by design and/or characterization production tested at  $T_J = T_A = 25^{\circ}\text{C}$ . Low duty cycle pulse techniques are used during testing to maintain the junction temperature as close to ambient as possible.
- A proper heatsinking and/or low duty cycle pulse techniques are used to operate the device within the Safe Operating Area.
- Guaranteed by design

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**Table 6. ELECTRICAL CHARACTERISTICS,  $V_{OUT} = 12\text{ V}$**  ( $-40^{\circ}\text{C} \leq T_J \leq 85^{\circ}\text{C}$ ;  $V_{IN} = 340\text{ V}$ ;  $I_{OUT} = 100\text{ }\mu\text{A}$ ,  $C_{IN} = 2.2\text{ }\mu\text{F}$ ,  $C_{OUT} = 22\text{ }\mu\text{F}$ , unless otherwise noted. Typical values are at  $T_J = +25^{\circ}\text{C}$ .) (Note 6)

| Parameter                             | Test Conditions                                                                                                                           | Symbol              | Min    | Typ  | Max    | Unit                |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|------|--------|---------------------|
| Operating Input Voltage DC            |                                                                                                                                           | $V_{IN}$            | 55     |      | 450    | V                   |
| Output Voltage Accuracy               | $T_J = 25^{\circ}\text{C}$ , $I_{OUT} = 100\text{ }\mu\text{A}$ , $55\text{ V} \leq V_{IN} \leq 450\text{ V}$                             | $V_{OUT}$           | 11.460 | 12   | 12.540 | V                   |
|                                       | $-40^{\circ}\text{C} \leq T_J \leq 85^{\circ}\text{C}$ , $I_{OUT} = 100\text{ }\mu\text{A}$ , $55\text{ V} \leq V_{IN} \leq 450\text{ V}$ | $V_{OUT}$           | 11.4   | 12   | 12.6   | V                   |
| Line Regulation                       | $55\text{ V} \leq V_{IN} \leq 450\text{ V}$ , $I_{OUT} = 100\text{ }\mu\text{A}$                                                          | $\text{Reg}_{LINE}$ | -0.5   | 0.1  | +0.5   | %/V                 |
| Load Regulation                       | $100\text{ }\mu\text{A} \leq I_{OUT} \leq 10\text{ mA}$ , $V_{IN} = 65\text{ V}$                                                          | $\text{Reg}_{LOAD}$ | -1.0   | 0.66 | +1.0   | %                   |
| Maximum Output Current (Note 7)       | $55\text{ V} \leq V_{IN} \leq 450\text{ V}$                                                                                               | $I_{OUT}$           | 10.5   |      |        | mA                  |
| Quiescent Current                     | $I_{OUT} = 0$ , $55\text{ V} \leq V_{IN} \leq 450\text{ V}$                                                                               | $I_Q$               |        | 17   | 22     | $\mu\text{A}$       |
| Ground Current (Note 7)               | $55\text{ V} \leq V_{IN} \leq 450\text{ V}$<br>$0 < I_{OUT} \leq 10\text{ mA}$                                                            | $I_{GND}$           |        |      | 25     | $\mu\text{A}$       |
| Power Supply Rejection Ratio          | $V_{IN} = 340\text{ V}_{DC} + 1\text{ V}_{pp}$ modulation, $I_{OUT} = 100\text{ }\mu\text{A}$                                             | $f = 1\text{ kHz}$  | PSRR   |      | 70     | dB                  |
| Noise                                 | $f = 100\text{ Hz to } 100\text{ kHz}$<br>$V_{IN} = 340\text{ V}_{DC}$ , $I_{OUT} = 100\text{ }\mu\text{A}$                               | $V_{NOISE}$         |        | 420  |        | $\mu\text{V}_{rms}$ |
| Thermal Shutdown Temperature (Note 8) | Temperature increasing from $T_J = +25^{\circ}\text{C}$                                                                                   | $T_{SD}$            |        | 145  |        | $^{\circ}\text{C}$  |
| Thermal Shutdown Hysteresis (Note 8)  | Temperature falling from $T_{SD}$                                                                                                         | $T_{SDH}$           | -      | 10   | -      | $^{\circ}\text{C}$  |

**Table 7. ELECTRICAL CHARACTERISTICS,  $V_{OUT} = 15\text{ V}$**  ( $-40^{\circ}\text{C} \leq T_J \leq 85^{\circ}\text{C}$ ;  $V_{IN} = 340\text{ V}$ ;  $I_{OUT} = 100\text{ }\mu\text{A}$ ,  $C_{IN} = 2.2\text{ }\mu\text{F}$ ,  $C_{OUT} = 22\text{ }\mu\text{F}$ , unless otherwise noted. Typical values are at  $T_J = +25^{\circ}\text{C}$ .) (Note 6)

| Parameter                             | Test Conditions                                                                                                                           | Symbol              | Min    | Typ  | Max    | Unit                |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------|------|--------|---------------------|
| Operating Input Voltage DC            |                                                                                                                                           | $V_{IN}$            | 60     |      | 450    | V                   |
| Output Voltage Accuracy               | $T_J = 25^{\circ}\text{C}$ , $I_{OUT} = 100\text{ }\mu\text{A}$ , $60\text{ V} \leq V_{IN} \leq 450\text{ V}$                             | $V_{OUT}$           | 14.325 | 15   | 15.675 | V                   |
|                                       | $-40^{\circ}\text{C} \leq T_J \leq 85^{\circ}\text{C}$ , $I_{OUT} = 100\text{ }\mu\text{A}$ , $60\text{ V} \leq V_{IN} \leq 450\text{ V}$ | $V_{OUT}$           | 14.25  | 15   | 15.75  | V                   |
| Line Regulation                       | $60\text{ V} \leq V_{IN} \leq 450\text{ V}$ , $I_{OUT} = 100\text{ }\mu\text{A}$                                                          | $\text{Reg}_{LINE}$ | -0.5   | 0.1  | +0.5   | %/V                 |
| Load Regulation                       | $100\text{ }\mu\text{A} \leq I_{OUT} \leq 10\text{ mA}$ , $V_{IN} = 65\text{ V}$                                                          | $\text{Reg}_{LOAD}$ | -1.0   | 0.66 | +1.0   | %                   |
| Maximum Output Current (Note 7)       | $65\text{ V} \leq V_{IN} \leq 450\text{ V}$                                                                                               | $I_{OUT}$           | 10.5   |      |        | mA                  |
| Quiescent Current                     | $I_{OUT} = 0$ , $60\text{ V} \leq V_{IN} \leq 450\text{ V}$                                                                               | $I_Q$               |        | 18   | 22     | $\mu\text{A}$       |
| Ground Current (Note 7)               | $60\text{ V} \leq V_{IN} \leq 450\text{ V}$<br>$0 < I_{OUT} \leq 10\text{ mA}$                                                            | $I_{GND}$           |        |      | 25     | $\mu\text{A}$       |
| Power Supply Rejection Ratio          | $V_{IN} = 340\text{ V}_{DC} + 1\text{ V}_{pp}$ modulation, $I_{OUT} = 100\text{ }\mu\text{A}$                                             | $f = 1\text{ kHz}$  | PSRR   |      | 70     | dB                  |
| Noise                                 | $f = 100\text{ Hz to } 100\text{ kHz}$<br>$V_{IN} = 340\text{ V}_{DC}$ , $I_{OUT} = 100\text{ }\mu\text{A}$                               | $V_{NOISE}$         |        | 500  |        | $\mu\text{V}_{rms}$ |
| Thermal Shutdown Temperature (Note 8) | Temperature increasing from $T_J = +25^{\circ}\text{C}$                                                                                   | $T_{SD}$            |        | 145  |        | $^{\circ}\text{C}$  |
| Thermal Shutdown Hysteresis (Note 8)  | Temperature falling from $T_{SD}$                                                                                                         | $T_{SDH}$           | -      | 10   | -      | $^{\circ}\text{C}$  |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

6. Performance guaranteed over the indicated operating temperature range by design and/or characterization production tested at  $T_J = T_A = 25^{\circ}\text{C}$ . Low duty cycle pulse techniques are used during testing to maintain the junction temperature as close to ambient as possible.

7. A proper heatsinking and/or low duty cycle pulse techniques are used to operate the device within the Safe Operating Area.

8. Guaranteed by design

TYPICAL CHARACTERISTICS

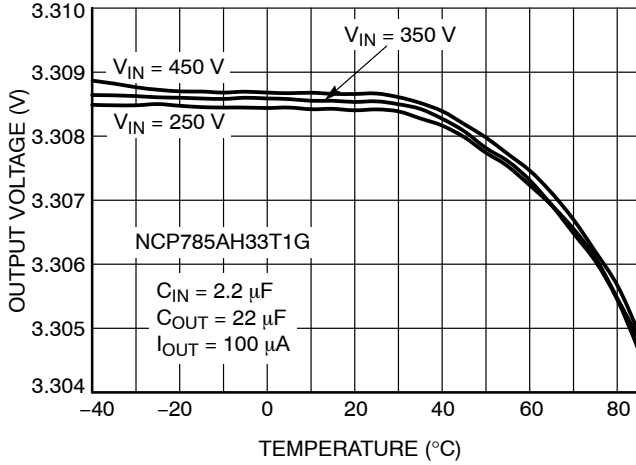


Figure 3. Output Voltage vs. Temperature

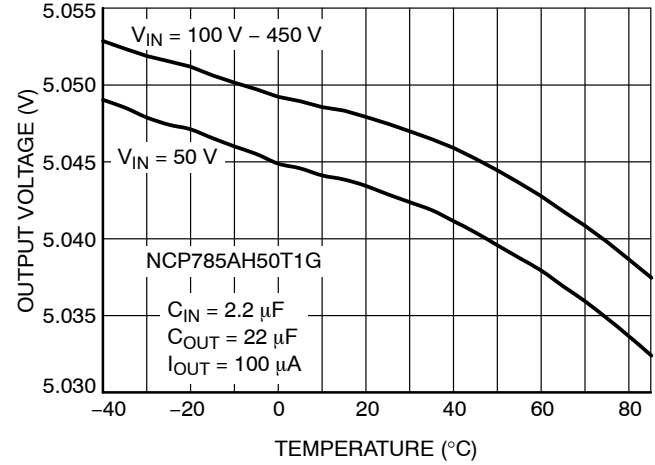


Figure 4. Output Voltage vs. Temperature

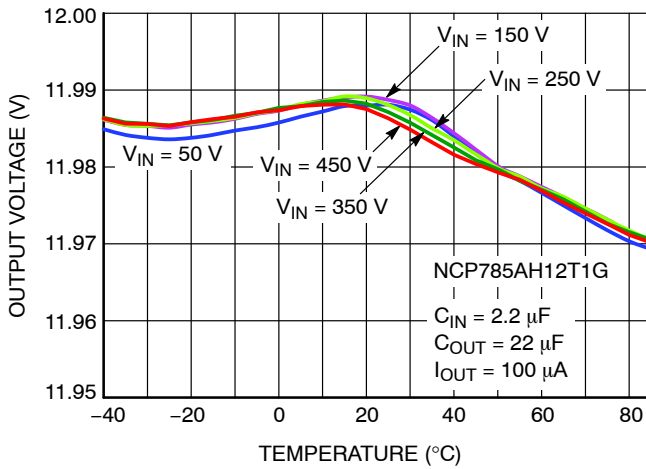


Figure 5. Output Voltage vs. Temperature

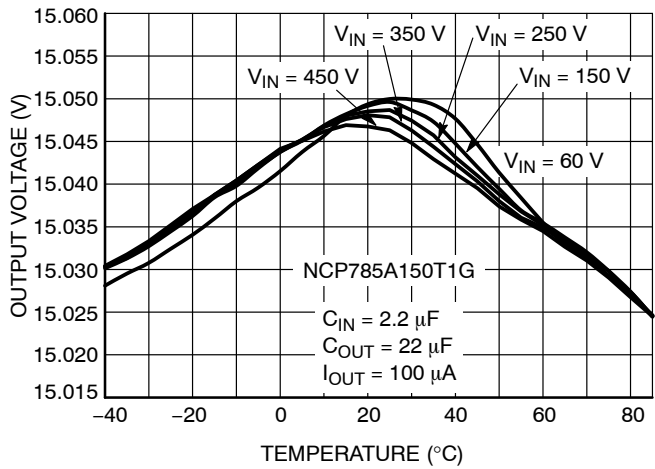


Figure 6. Output Voltage vs. Temperature

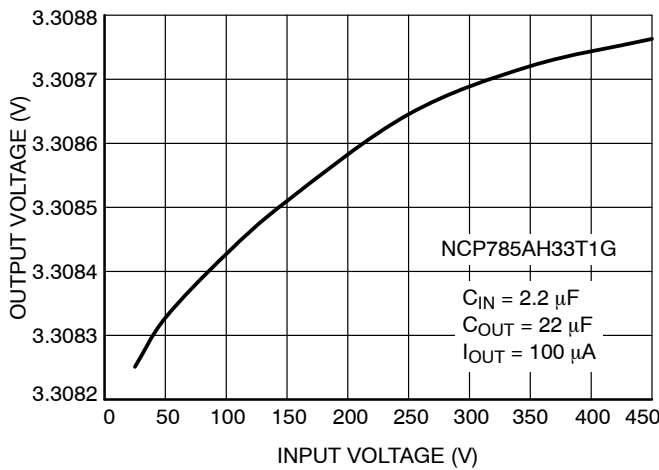


Figure 7. Output Voltage vs. Input Voltage

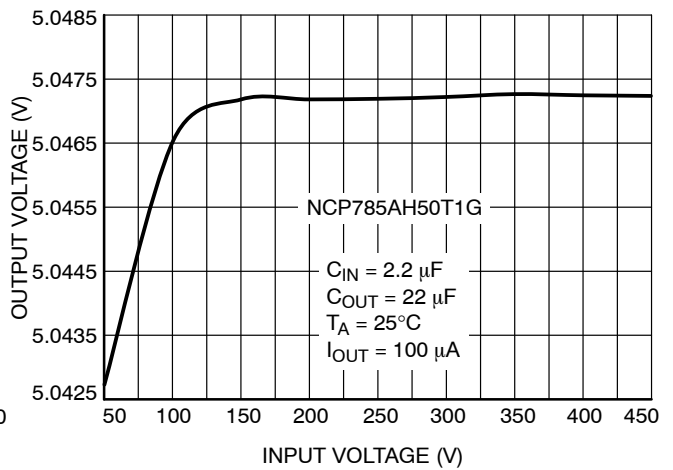


Figure 8. Output Voltage vs. Input Voltage

TYPICAL CHARACTERISTICS

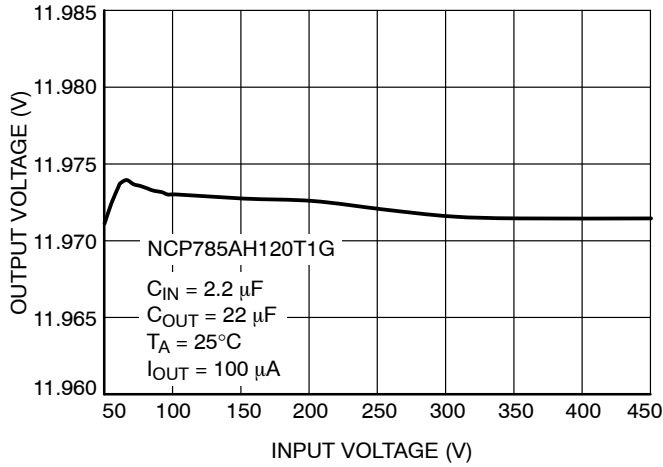


Figure 9. Output Voltage vs. Input Voltage

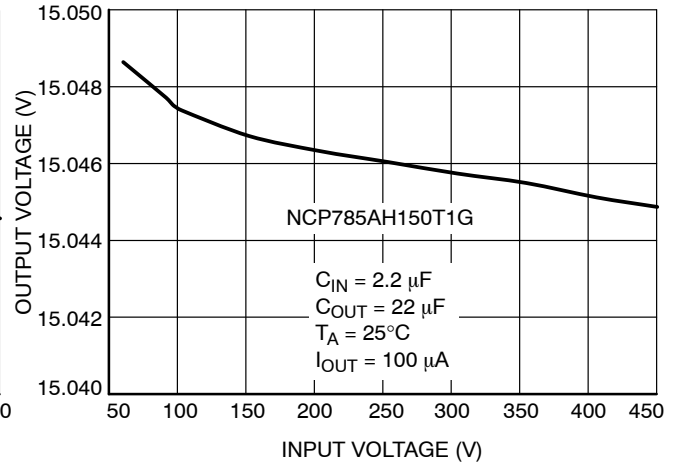


Figure 10. Output Voltage vs. Input Voltage

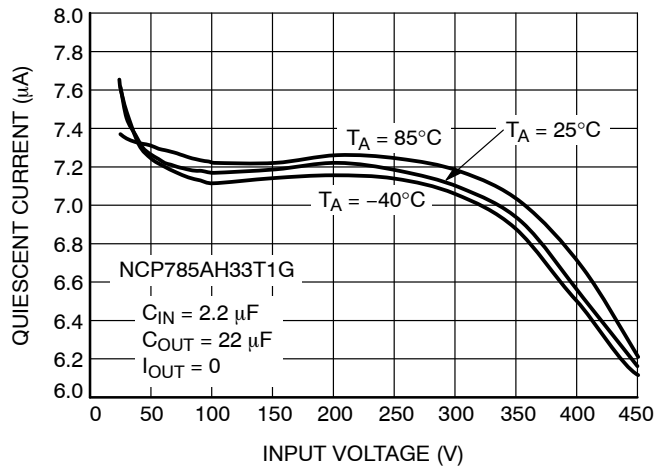


Figure 11. Quiescent Current vs. Input Voltage

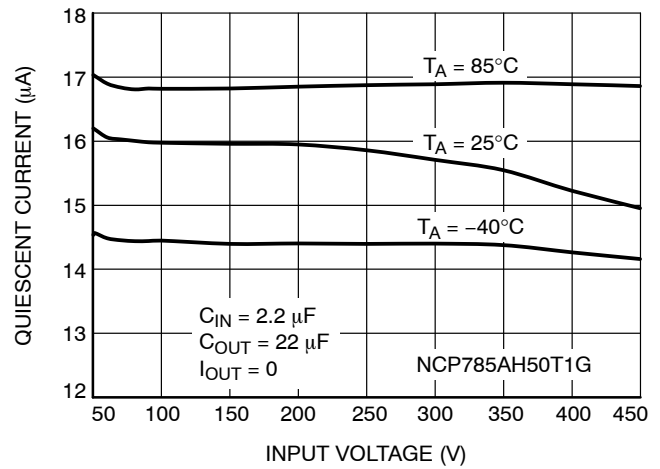


Figure 12. Quiescent Current vs. Input Voltage

TYPICAL CHARACTERISTICS

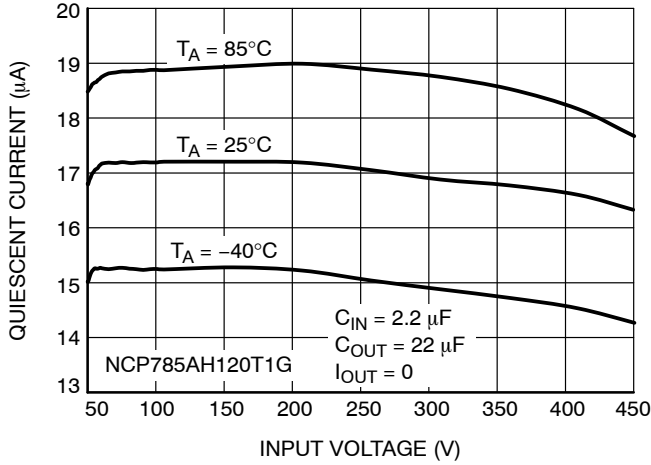


Figure 13. Quiescent Current vs. Input Voltage

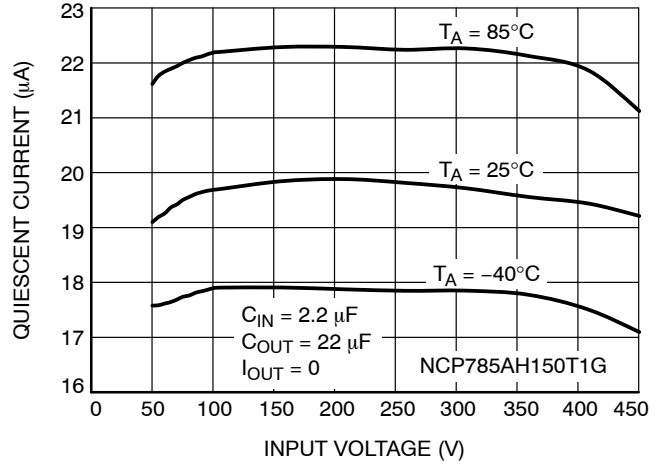


Figure 14. Quiescent Current vs. Input Voltage

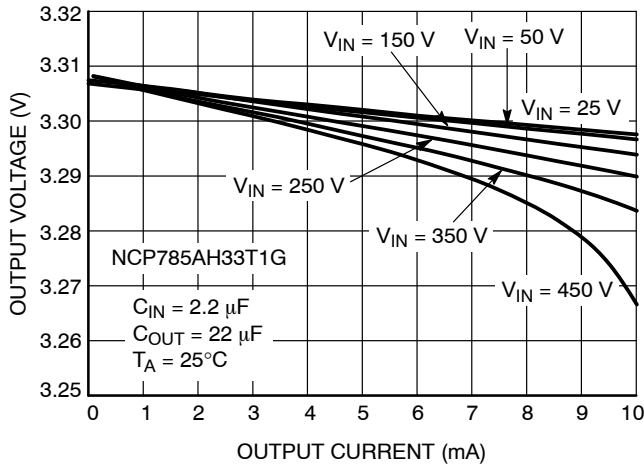


Figure 15. Output Voltage vs. Output Current

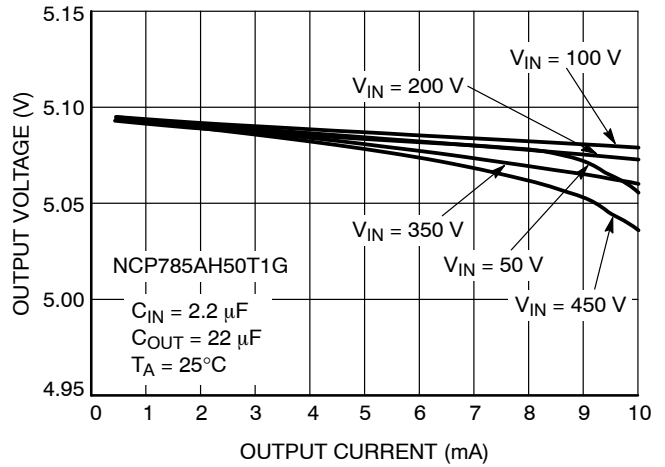


Figure 16. Output Voltage vs. Output Current

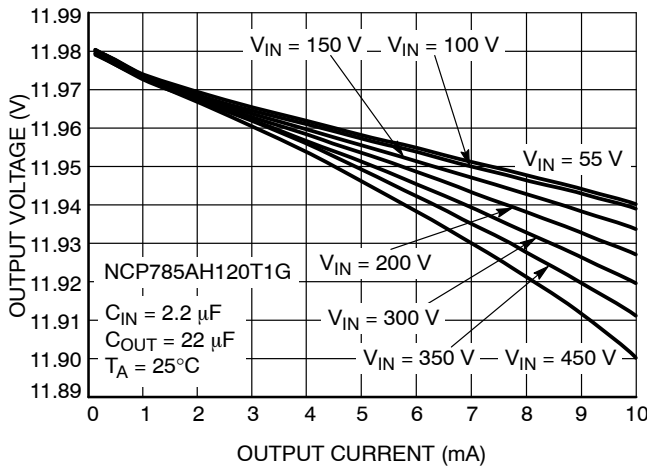


Figure 17. Output Voltage vs. Output Current

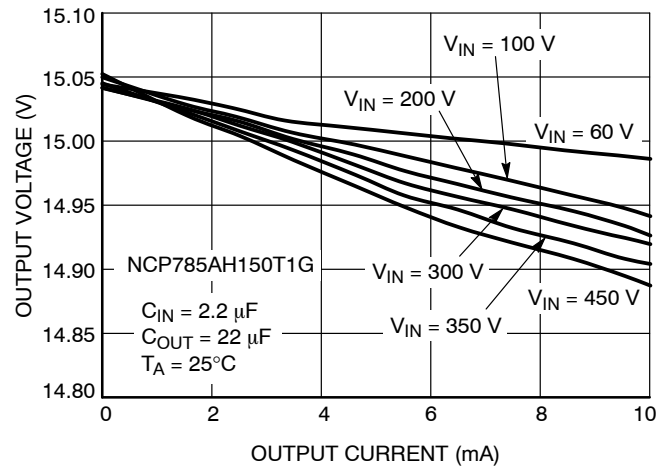


Figure 18. Output Voltage vs. Output Current

TYPICAL CHARACTERISTICS

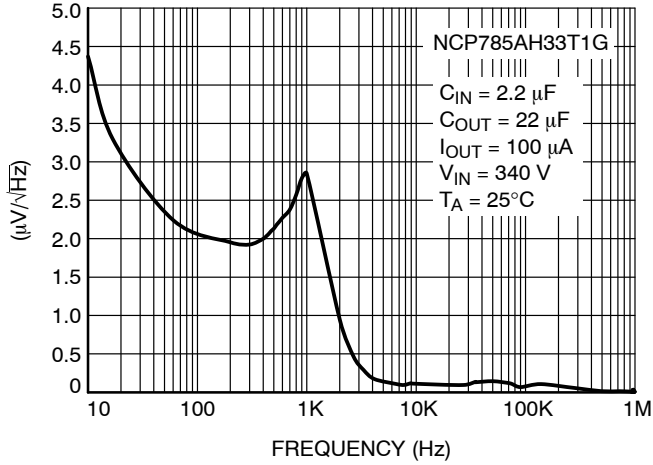


Figure 19. Output Noise Density vs. Frequency

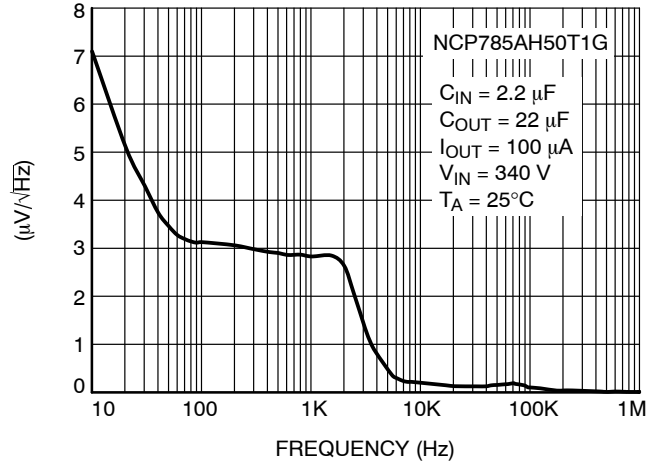


Figure 20. Output Noise Density vs. Frequency

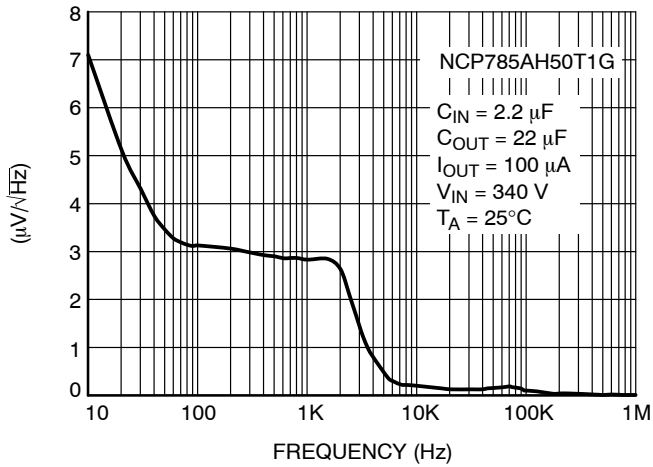


Figure 21. Output Noise Density vs. Frequency

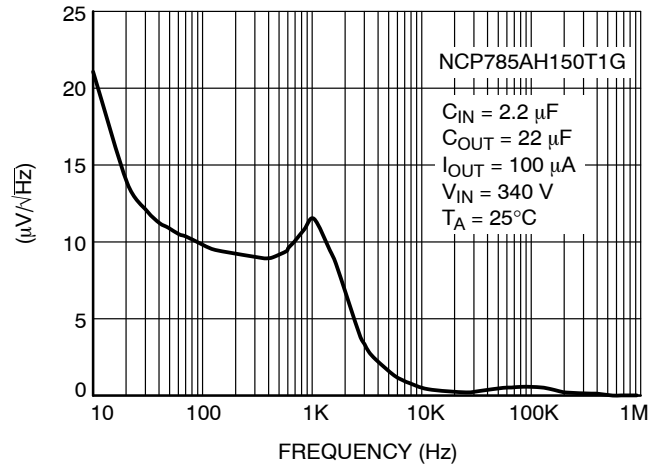


Figure 22. Output Noise Density vs. Frequency



# NCP785A

## APPLICATION INFORMATION

The typical application circuit for the NCP785A device is shown below.

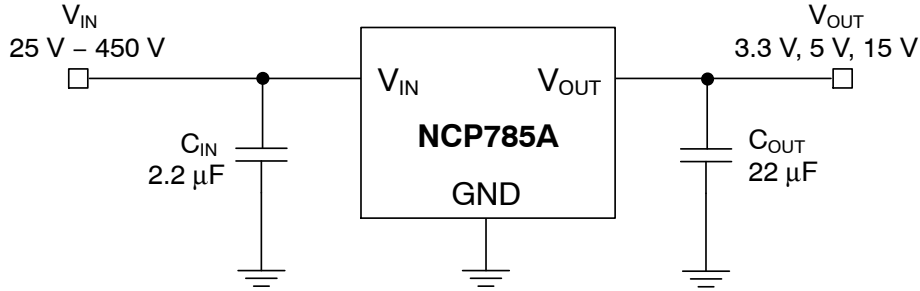


Figure 23. Typical Application Schematic

### Input Decoupling (C1)

A 1  $\mu\text{F}$  capacitor either ceramic or electrolytic is recommended and should be connected close to the input pin of NCP785A. Higher value 2.2  $\mu\text{F}$  is necessary to sustain the required minimum input voltage at full load for AC voltage as low as 85 V with half wave rectifier.

### Output Decoupling (C2)

The NCP785A Regulator does not require any specific Equivalent Series Resistance (ESR). Thus capacitors exhibiting ESRs ranging from a few  $\text{m}\Omega$  up to 0.5  $\Omega$  can be used safely. The minimum decoupling value is 22  $\mu\text{F}$ . The regulator accepts ceramic chip capacitors as well as tantalum devices or low ESR electrolytic capacitors. Larger values improve noise rejection and load transient response.

### Layout Recommendations

Please be sure the  $V_{\text{IN}}$  and GND lines are sufficiently wide. When the impedance of these lines is high, there is a chance to pick up noise or to cause the malfunction of regulator.

Set external components, especially the output capacitor, as close as possible to the circuit, and make leads as short as possible.

### Thermal

As power across the NCP785A increases, it might become necessary to provide some thermal relief. The maximum power dissipation supported by the device is dependent upon board design layout and used package. Mounting pad configuration on the PCB, the board material, and also the ambient temperature affect the rate of temperature rise for the part. This is stating that when the NCP785A has good thermal conductivity through the PCB, the junction temperature will be relatively low with high power dissipation applications.

## ORDERING INFORMATION

| Part Number    | Output Voltage | Marking | Package             | Shipping <sup>†</sup> |
|----------------|----------------|---------|---------------------|-----------------------|
| NCP785AH33T1G  | 3.3 V          | AA      | SOT-89<br>(Pb-Free) | 1000 / Tape & Reel    |
| NCP785AH50T1G  | 5 V            | AC      |                     |                       |
| NCP785AH120T1G | 12 V           | AJ      |                     |                       |
| NCP785AH150T1G | 15 V           | AD      |                     |                       |

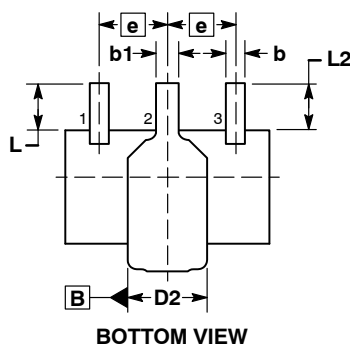
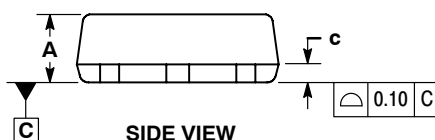
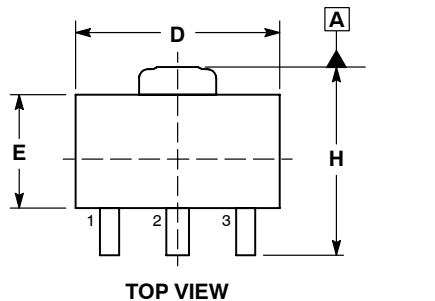
<sup>†</sup>For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.



SCALE 2:1

**SOT-89, 3 LEAD**  
**CASE 528AG**  
**ISSUE O**

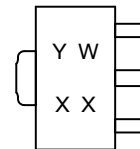
DATE 04 MAR 2014



## NOTES:

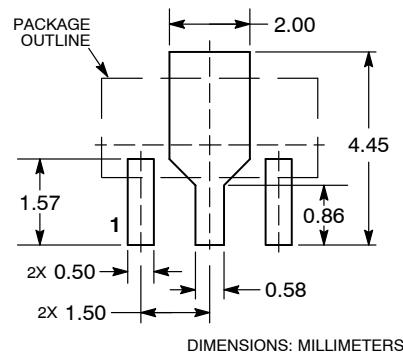
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 1994.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. LEAD THICKNESS INCLUDES LEAD FINISH.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH, PROTRUSIONS, OR GATE BURRS.
5. DIMENSIONS L, L2, D2, AND H ARE MEASURED AT DATUM PLANE C.
6. CENTER LEAD CONTOUR MAY VARY WITHIN THE REGION DEFINED BY DIMENSION E.
7. DIMENSION D2 IS DEFINED AT ITS WIDEST POINT.

| DIM | MILLIMETERS |      |
|-----|-------------|------|
|     | MIN         | MAX  |
| A   | 1.40        | 1.60 |
| b   | 0.38        | 0.47 |
| b1  | 0.46        | 0.55 |
| c   | 0.40        | 0.44 |
| D   | 4.40        | 4.60 |
| D2  | 1.60        | 1.90 |
| E   | 2.40        | 2.60 |
| e   | 1.50 BSC    |      |
| H   | 4.05        | 4.25 |
| L   | 0.89        | 1.20 |

**GENERIC**  
**MARKING DIAGRAM\***


Y = Year  
W = Work Week  
XX = Specific Device Code

\*This information is generic. Please refer to device data sheet for actual part marking. Pb-Free indicator, "G" or microdot "▪", may or may not be present. Some products may not follow the Generic Marking.

**RECOMMENDED**  
**MOUNTING FOOTPRINT\***


\*For additional information on our Pb-Free strategy and soldering details, please download the ON Semiconductor Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.

|                         |                       |                                                                                                                                                                                  |
|-------------------------|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
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